

SKIN DISEASE DETECTION USING PYTHON AND DEEP LEARNING



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ABSTRACT

- Skin diseases like Eczema, Psoriasis, Melanoma and Ringworm are common worldwide and require early diagnosis to prevent serious complications. This project focuses on developing an automated skin disease detection system using Python and Deep Learning techniques. The system analyzes skin images and classifies diseases using Convolutional Neural Networks (CNN). A labeled dataset is used for training, and preprocessing methods such as resizing, noise reduction, and normalization are applied to improve accuracy. Important features are extracted from images, and the trained model predicts the type of disease. The system is tested with new input images to ensure reliability. It provides a user-friendly interface where users can upload images and receive predictions easily. This approach supports early detection, reduces dependency on dermatologists, and improves accessible healthcare.

INTRODUCTION

- Skin diseases are common and can range from minor infections to serious conditions, making early detection important to prevent complications and ensure proper treatment. However, lack of access to dermatologists, especially in rural areas, often leads to delays in diagnosis. With advancements in Artificial Intelligence and Deep Learning, medical image analysis has become more efficient and accurate. This project uses Python and Convolutional Neural Networks (CNN) to detect skin diseases from images. The system analyzes skin images and classifies different diseases with good accuracy. It provides a simple platform for users to upload images and get predictions. Overall, it helps in early detection and improves access to healthcare.

LITERATURE REVIEW

S.NO	YEAR OF PUBLICATION	TITLE	AUTHORS	TECHNIQUES	ACCURACY	LIMITATIONS
1.	2025	A Review of Skin Disease Detection Using Deep Learning .	Shereen Fatima Bhatti, Hidayatullah Shaikh, Attaullah Sahito	CNN, Mobile Net, Efficient Net, Vision Transformers, Explainable AI.	85%	Dataset imbalance, lack of diversity, computational cost
2.	2024	Deep Learning Models Across the Range of Skin Disease	Kaushik P. Venkatesh et al.	CNNs, multimodal dermatology AI models.	~80–90%	Limited real-world validation, generalization issues.

LITERATURE SURVEY

S.N O	YEAR OF PUBLICATION	TITLE	AUTHORS	TECHNIQUES	ACCURACY	LIMITATIONS
3.	2023	Systematic Review of Deep Learning Image Analyses for Skin Disease.	Shern Ping Choy et al.	CNN-based classification & severity grading.	89–94% (varies by disease)	High bias risk, lack of external validation datasets.
4.	2023	Skin Cancer Detection Using Deep Learning—A Review	Maryam Naqvi, Syed Qasim Gilani et al.	CNN, segmentation, classification models	~90%	Relies heavily on dermoscopic images; not generalizable.

S.NO	YEAR OF PUBLICATION	TITLE	AUTHORS	TECHNIQUES	ACCURACY	LIMITATIONS
5.	2023	Web-based Mpox Skin Lesion Detection System	Shams Nafisa Ali et al.	DenseNet121, MobileNetV2, Efficient Net, Transfer Learning	83.5%	Dataset bias, needs more diverse skin tone representation.
6.	2022	Monkeypox Skin Lesion Detection Using Deep Learning Models	Monkeypox Skin Lesion Detection Using Deep Learning Models	VGG16, ResNet50, InceptionV3, Ensemble model	82.9%	Small dataset, poor diversity, limited real-world testing.
7.	2021	Skin Disease Diagnosis with Deep Learning: A Review	Hongfeng Li et al.	CNN architectures, image preprocessing, DL frameworks	~80–90%	Requires large datasets, overfitting issues.

S.NO	YEAR OF PUBLICATION	TITLE	AUTHORS	TECHNIQUES	ACCURACY	LIMITATIONS
8.	2019	Deep Learning System for Differential Diagnosis of Skin Diseases	Yuan Liu et al.	Deep Neural Networks (DNN), multi-class classification	Top-1:71%, Top-3: 93%.	Needs clinical validation, dataset bias

PROBLEM STATEMENT

Skin diseases are increasingly common and can lead to serious health complications if not detected at an early stage. However, accurate diagnosis requires expert dermatologists, and access to such medical professionals is limited, especially in rural and remote areas. Traditional diagnosis methods are time-consuming and may not always be readily available to everyone. There is a need for an efficient, cost-effective, and automated system that can analyze skin images and identify diseases accurately. Therefore, this project aims to develop a skin disease detection system using Python and Deep Learning techniques to provide quick, reliable, and accessible diagnosis support.

METHODOLOGY



EXPLANATION FOR METHODOLOGY

1. **DATA COLLECTION:** The dataset of skin disease images is collected from reliable sources such as Kaggle. It includes different types of diseases to ensure proper training of the model.
2. **DATA PREPARATION:** The collected data is organized into categories and cleaned by removing unwanted or duplicate images. This step ensures consistency and better performance during training.
3. **IMAGE PROCESSING:** Images are resized to a fixed dimension and processed using techniques like noise reduction and normalization. This improves the quality of images for accurate analysis.
4. **FEATURE EXTRACTION:** Important features such as color, texture, and patterns are extracted from the images. These features help the model to distinguish between different skin diseases.
5. **MODEL TRAINING:** The system is trained using Python code and a Convolutional Neural Network (CNN) model. The model learns patterns from the dataset to improve prediction accuracy.
6. **CLASSIFICATION:** The trained model classifies the input image into a specific type of skin disease. It predicts the disease based on learned features from training data.
7. **OUTPUT GENERATION:** The output is generated showing the predicted type of disease. The system also displays accuracy to observe the performance of the model.
8. **RESULT AND MONITORING:** The results are analyzed based on prediction accuracy and disease classification. Continuous monitoring is done to improve system reliability and performance.

PROGRAM



skin_disease.ipynb ☆ ☁

File Edit View Insert Runtime Tools Help



Share



Q Commands + Code + Text ▶ Run all

✓ RAM Disk



Files



..

sample_data

README.md

anscombe.json

california_housing_test.csv

california_housing_train.csv

mnist_test.csv

mnist_train_small.csv

archive.zip

Disk 87.70 GB available

[]

```
!file archive.zip
```

▼

archive.zip: Zip archive data, at least v4.5 to extract, compression method=deflate

[]

```
!ls
```

▼

1.jpg archive.zip sample_data

[]

```
!file archive.zip
```

▼

archive.zip: Zip archive data, at least v4.5 to extract, compression method=deflate

[]



```
!unzip archive.zip -d skin_dataset
```

>

... [Show hidden output](#)

[]

```
import os
print(os.listdir("/content"))
```

{ } Variables Terminal



✓ 12:14 AM Python 3

OUTPUT

```
from google.colab import files

uploaded = files.upload()
```

... Choose Files No file chosen Cancel upload

```
filename = list(uploaded.keys())[0]
img = image.load_img(filename, target_size=(224,224))
plt.imshow(img)
plt.axis("off")
plt.show()
...
```



```
img_array = image.img_to_array(img)
img_array = np.expand_dims(img_array, axis=0)

prediction = model.predict(img_array)

predicted_class = class_names[np.argmax(prediction)]
confidence = np.max(prediction) * 100

print("Predicted Disease :", predicted_class)
print("Confidence : {:.2f}%".format(confidence))
```

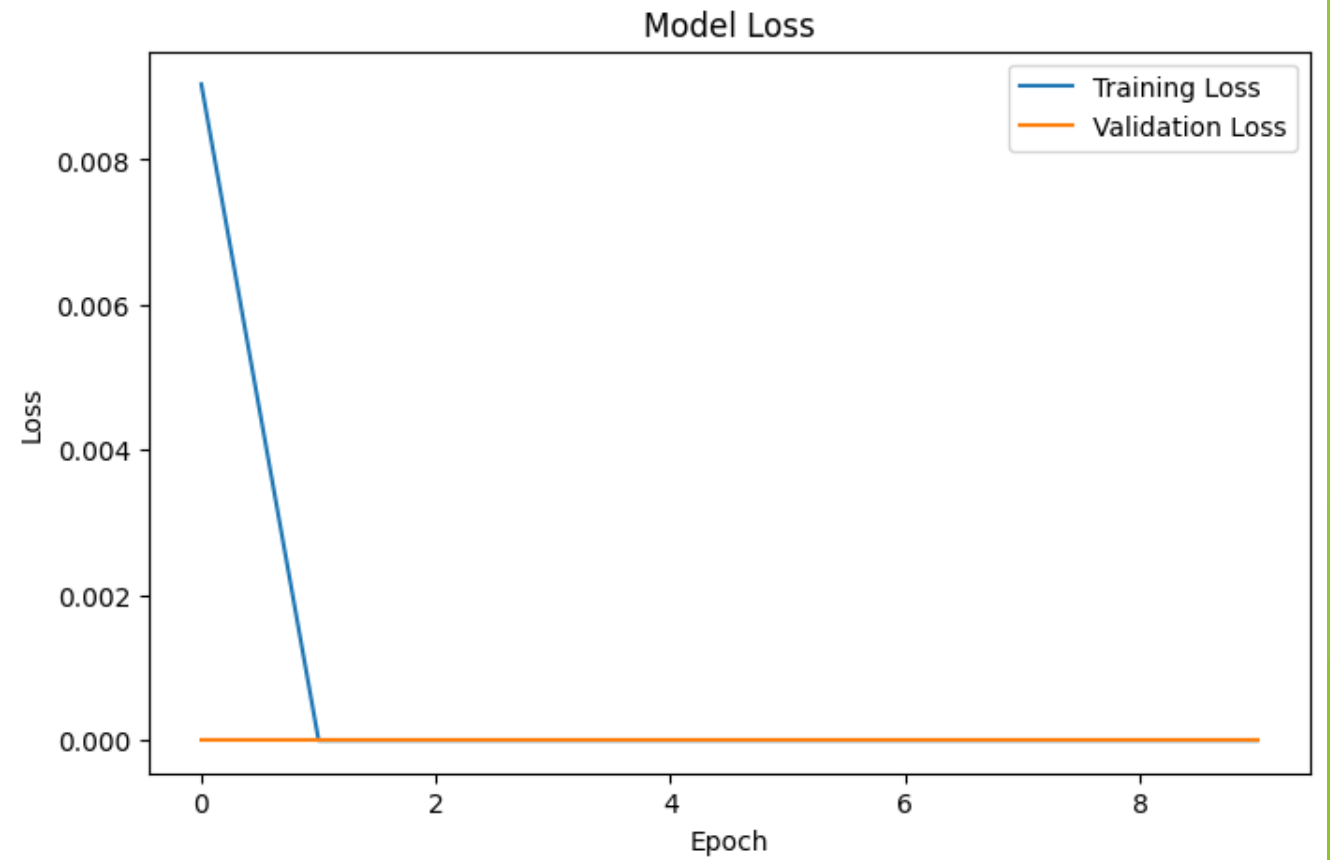
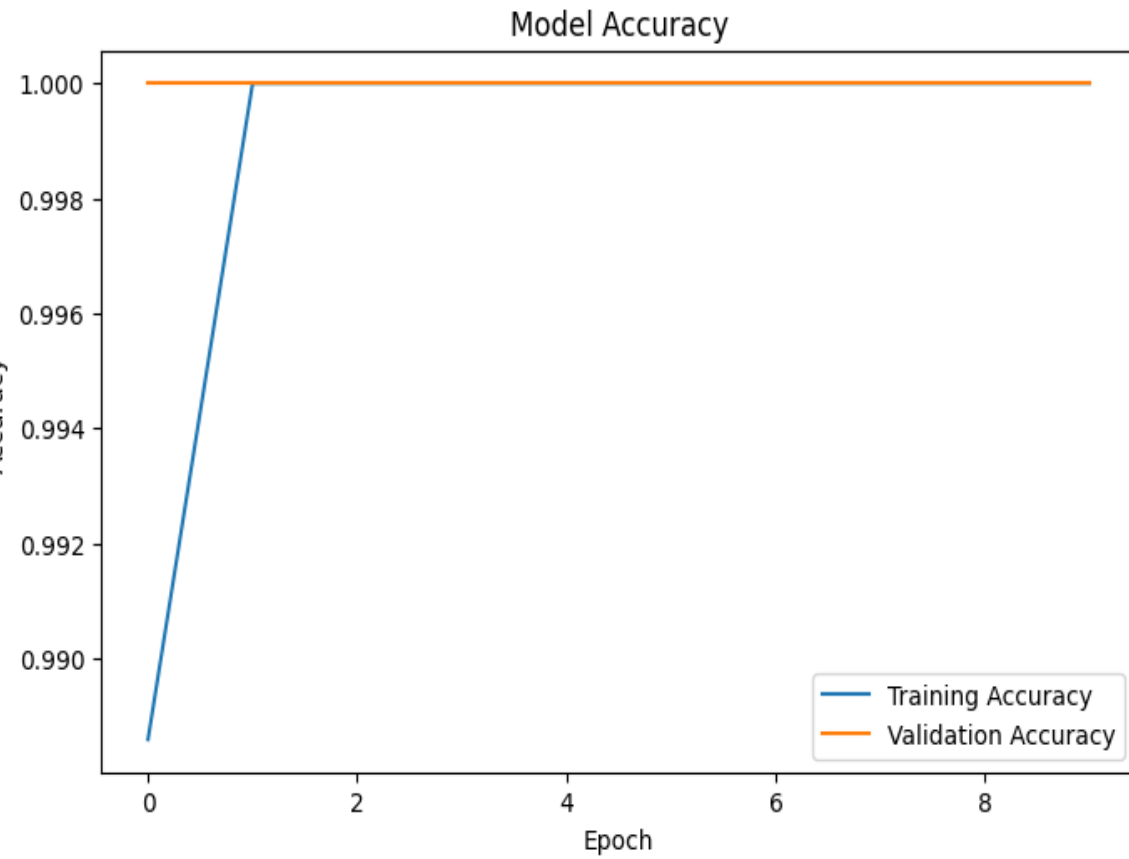
... 1/1 ————— 12s 12s/step
Predicted Disease : Eczema
Confidence : 80.35%

The user uploads a skin disease image through the Google Colab interface for analysis.

- The uploaded skin image is displayed for visual verification before processing.

- The predicted disease name and confidence percentage are displayed as the final output.

GRAPHICAL REPRESENTATION



FUTURE ENHANCEMENT



- Improve model accuracy using larger and more diverse datasets.
- Detect a wider range of skin diseases.
- Develop a mobile application for real-time skin disease detection.
- Integrate cloud-based storage and processing.
- Provide treatment recommendations and dermatologist consultation support.
- Support multiple languages for better accessibility.
- Enhance prediction speed and performance using advanced deep learning models.

CONCLUSION

The proposed system successfully detects and classifies skin diseases using Python and Deep Learning techniques. It provides accurate and quick predictions, helping in early diagnosis and reducing dependency on dermatologists. The system improves healthcare accessibility and can be further developed into real-time or mobile-based applications. It uses image processing and CNN models to achieve reliable results. The model is trained and tested using labeled datasets to ensure better performance. This approach helps in saving time and cost for patients. In future, the system can be further developed by increasing dataset size, improving model accuracy, and integrating it with mobile or web applications for real-time usage.

THANK YOU